Measures of central tendency

Measures of central tendency are statistical measures that describe the center of a dataset. The three main measures are mean, median, and mode.

- mean → average
- median → middle value
- mode → highest frequency

numeric data

Age → [45, 74, 47, 52, 65, 64, 43]

$$\text{mean} \rightarrow \frac{45+74+47+52+65+64+43}{7}$$

↳ average of age

↳ sort the data in an ascending order

$n = 7$ (odd)

43, 45, 47, 52, 64, 65, 74

median = 52

(Assumption)

$n = 8$ (even)

43, 45, 47, $\boxed{52, 64}$, 65, 74, 80

$$\text{median} = \frac{52+64}{2}$$

Mode → Categorical data

Gender → [M, f, F, M, f, f, M, M]

$\longrightarrow$ M

$\longrightarrow$ f

Outliers $\longrightarrow$ extreme rare points in our data

As a Data Analyst

{60K, 50K, 40K, 90K, 45K, 10K, 1 Cr}

outliers are available

↳ box plot

average salary of people as a fresher earning in Bengaluru (DA)

$$\frac{(60K + 50K + 40K + 90K + 45K + 10K + 1Cr)}{7}$$

Median is a robust measure when Outliers are available in the data

extreme points

{10K, 40K, 45K, $\boxed{50K}$, 60K, 90K, 1 Cr}

Median